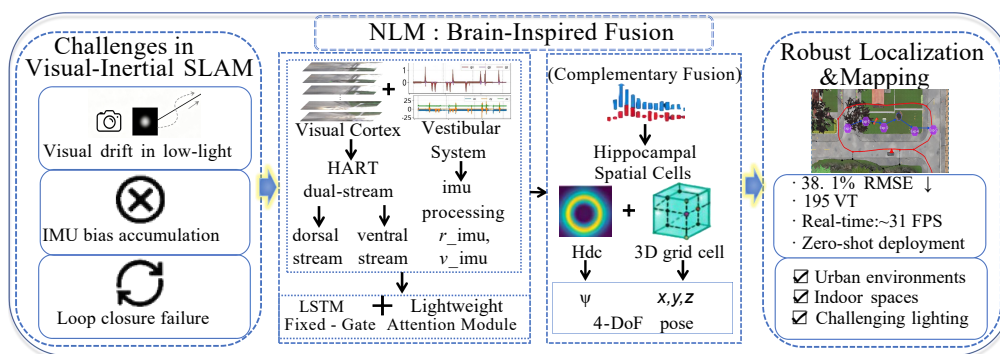


Graphical Abstract

Brain-Inspired 4-DoF Localization-Mapping: Complementary Vestibular-Visual Fusion with Spatial Cells



Highlights

Brain-Inspired 4-DoF Localization-Mapping: Complementary Vestibular-Visual Fusion with Spatial Cells

- A brain-inspired 4-DoF localization and mapping framework integrating 3-D grid cells and complementary vestibular-visual multimodal fusion
- 39-fold expansion of visual scene templates (from 5 to 195) to enable fine-grained, robust loop closure detection
- Achieves an average 38.1% RMSE reduction compared with EKF fusion baseline over CARLA, EuRoC and KITTI datasets
- Training-free real-time implementation at 31 FPS with strong zero-shot cross-domain generalization across urban, highway and indoor environments

Brain-Inspired 4-DoF Localization-Mapping: Complementary Vestibular-Visual Fusion with Spatial Cells

Abstract

Brain-inspired spatial cognition offers interpretable alternatives to opaque learning methods for autonomous localization. However, existing brain-inspired frameworks remain limited to 2-D environments or simplified visual processing, lacking robust 3-D spatial representations under challenging conditions. To address these limitations, this paper proposes NLM, a brain-inspired localization-mapping framework extended to 4-DoF state estimation within 3-D space, integrating vestibular-visual complementary fusion with hippocampal-entorhinal spatial representations. The framework comprises four core modules: dual-stream visual processing (ventral for recognition, dorsal for motion), vestibular IMU modeling, 3-D grid cells with face-centered cubic lattice for position encoding, and experience-based cognitive mapping for loop closure. Unlike data-driven approaches, our brain-inspired design achieves real-time performance (~ 31 FPS on CPU) with zero-shot deployment capability. Experiments on CARLA (Town 01 / 02 / 10) demonstrate 38.1% average RMSE improvement over EKF baselines, validating the framework’s effectiveness for autonomous localization and motion across diverse environments.

Keywords: Brain-inspired localization-mapping, Vestibular-visual fusion, Hippocampal spatial cells, Visual place recognition, Cognitive mapping

1. Introduction

Spatial localization and mapping in complex 3-D environments is fundamental to autonomous driving, mobile robotics, and intelligent transportation systems [1]. As the core sensing module of autonomous devices, SLAM directly supports all autonomous movement planning and execution. Modern practical approaches typically combine cameras, IMUs, and sometimes GNSS

to achieve robust pose estimation [2]. Conventional SLAM solutions mostly adopt optimization or filtering frameworks for pose solving, but they commonly have obvious defects in model interpretability and computing resource consumption.

Interestingly, organisms such as bats complete 3-D navigation by fusing visual and vestibular signals with hippocampal spatial cells [3, 4, 5, 6, 7]. These neural mechanisms enable brain-inspired localization frameworks to outperform traditional SLAM in interpretability and computational efficiency.

However, translating these biological principles into practical robotic systems faces several challenges. First, most existing bio-inspired frameworks operate in 2-D, whereas real-world deployments require robust 3-D spatial representation. Second, classical visual template matching is sensitive to lighting and viewpoint variations, limiting place recognition performance in challenging environments. Third, a simple yet effective fusion strategy that combines visual odometry with inertial measurements—without relying on computationally expensive filters like EKF—remains under-explored for brain-inspired localization frameworks.

To address these challenges, we present NLM, a brain-inspired localization-mapping framework extended to 4-DoF pose estimation within 3-D space (pose x, y, z, ψ) that integrates a dual-stream visual front-end (ventral pathway for scene recognition, dorsal pathway for motion extraction) and a vestibular front-end for IMU processing, with complementary filtering for multi-sensory fusion. The framework employs 3-D grid cells and multilayered head direction cells for spatial representation, and a multilayer experience map functioning as hippocampal-like episodic memory for place recognition and drift correction. For ground vehicles operating on relatively flat terrain (e.g., urban roads, indoor floors), the orientation is dominated by yaw angle ψ , while pitch and roll variations are typically constrained by vehicle dynamics ($< 5^\circ$). Our 4-DoF representation provides efficient localization-mapping for such scenarios [8, 9].

The contributions of this paper are summarized as follows:

- We propose a unified brain-inspired architecture that couples dual-stream visual processing with vestibular modeling and hippocampal-entorhinal spatial representations for 4-DoF localization and mapping.
- We develop cortex-inspired visual processing with explicit operators for visual template creation, increasing template count from 5 (typical in

RatSLAM) to 195 ($39 \times$ improvement) for finer-grained place discrimination under challenging lighting and viewpoint variations.

- We design a lightweight complementary fusion strategy for vestibular-visual integration that operates without computationally expensive filters, achieving real-time performance (~ 31 FPS on CPU) without requiring training data.
- We validate the framework on six diverse scenarios (CARLA Town 01 / 02 / 10, EuRoC MH_01 / 03, KITTI 07), achieving an average RMSE improvement of 38.1% over EKF baselines across all tested environments.

2. Related Work

We have summarized research work on different aspects of brain-inspired localization and mapping systems.

2.1. Visual SLAM and Sensor Fusion

Visual SLAM has achieved significant progress over the past two decades. Mur-Artal et al. proposed ORB-SLAM [10], which extracts sparse ORB keypoints for robust real-time pose estimation and has become a widely-used baseline in the field. Engel et al. developed LSD-SLAM [11] and DSO [12], which operate directly on pixel intensities without feature extraction, achieving semi-dense reconstruction in textured environments. For loop closure detection, Cummins and Newman proposed FAB-MAP [13], a probabilistic approach using bag-of-words models, while Arandjelovic et al. introduced NetVLAD [14], which learns place descriptors end-to-end for robust place recognition. However, these methods face critical limitations: ORB-SLAM requires sufficient texture (typically >100 features per frame) and fails in feature-poor environments, while direct methods like DSO struggle with motion blur and require photometric calibration. Both approaches incur substantial computational cost—ORB-SLAM’s feature extraction and matching dominate processing time, while DSO’s dense photometric optimization limits real-time performance on embedded platforms.

Integrating IMUs with cameras addresses visual SLAM limitations by providing lighting-invariant motion estimates. Mourikis and Roumeliotis proposed MSCKF [15], a tightly-coupled visual-inertial estimator achieving high accuracy through multi-state constraint optimization. Qin et al. developed

VINS-Mono [16], which combines tightly-coupled optimization with loop closure for robust monocular visual-inertial odometry. Mahony et al. [17] and Madgwick et al. [18] proposed complementary filtering approaches that exploit frequency-domain sensor characteristics—visual odometry provides accurate low-frequency estimates while IMU captures high-frequency dynamics. However, tightly-coupled methods like MSCKF and VINS-Mono require careful initialization of IMU biases and scale, involve complex Jacobian derivations, and lack biological interpretability. To address these limitations, we adopt a bio-inspired complementary filtering strategy that mimics biological vestibular-visual integration [19, 20], providing interpretable sensor fusion with minimal parameter tuning and avoiding the computational burden of joint nonlinear optimization.

2.2. Brain-Inspired Spatial Cognition

Biological navigation systems achieve robust spatial cognition through specialized neural circuits. O’Keefe and Dostrovsky discovered place cells in the hippocampus [21], which fire at specific locations forming cognitive maps. Hafting et al. discovered grid cells in the entorhinal cortex [22], exhibiting periodic hexagonal firing patterns that provide metric path integration. Taube et al. identified head direction cells [23], which encode facing direction like an internal compass. Moser et al. [24] provided comprehensive reviews of how these spatial cells work together for navigation. Recent discoveries by Finkelstein et al. demonstrated that bats possess 3-D place cells, 3-D grid cells with face-centered cubic (FCC) lattice structure, and 3-D head direction cells [25, 6], providing biological grounding for volumetric spatial representations.

Milford et al. pioneered RatSLAM [8, 26], translating rodent spatial cognition to robotics by implementing continuous attractor networks for pose cell dynamics and experience maps for loop closure, successfully demonstrating suburb-scale mapping up to 66 km. Steckel and Peremans developed BatSLAM [27], extending bio-inspired SLAM to sonar sensing. Yu et al. proposed NeuroSLAM (unpublished work), extending to 3-D position encoding with CNN-based visual processing. Lowry et al. [28] analyzed the metric-topological duality in spatial maps: metric maps provide geometric precision but scale poorly, while topological maps enable robust loop closure but lack metric accuracy. However, existing bio-inspired frameworks face three key limitations: (1) most operate in 2D or encode only 3-D position without orientation, lacking the 4-DoF representation needed for aerial/underwater

navigation; (2) visual front-ends remain simplistic (e.g., RatSLAM’s 1D intensity scanlines yield only ~ 5 templates in complex environments); and (3) IMU integration is absent or ad-hoc, missing the principled vestibular-visual fusion observed in biological systems. To address these gaps, we implement 3-D grid cells with FCC lattice structure and multilayered head direction cells for full 4-DoF spatial representation, integrate a dual-stream visual processing pipeline that increases template discrimination by $39\times$, and adopt bio-inspired complementary filtering for vestibular-visual integration.

2.3. Learning-Based Visual Processing

Deep learning offers powerful tools for visual place recognition. Arandjelovic et al. proposed NetVLAD [14], introducing differentiable VLAD layers for end-to-end place recognition training. Goodale and Milner [29] and Mishkin et al. [30] established the dual-stream theory of biological vision: the ventral stream ($V1 \rightarrow V4 \rightarrow IT$) processes object identity and scene recognition, while the dorsal stream ($V1 \rightarrow MT \rightarrow MST$) processes spatial relationships and motion [31, 32]. Vaswani et al. proposed Transformer architectures [33], which capture long-range dependencies through self-attention mechanisms, enabling effective temporal modeling for sequence-based tasks. However, learning-based methods face deployment challenges: NetVLAD requires large-scale training datasets (e.g., Google Street View with millions of images) and struggles to generalize to novel environments without retraining; standard Transformers incur quadratic complexity $\mathcal{O}(N^2)$ in sequence length for full self-attention, limiting real-time applicability; and learned representations lack explicit spatial structure, hindering interpretability and integration with metric mapping. To overcome these limitations, we adopt a training-free approach combining dual-stream visual processing inspired by biological ventral/dorsal pathways with a lightweight feature enhancement mechanism that captures global-local interactions through statistical modulation rather than full attention matrices. This design increases visual template discrimination from 5 (RatSLAM) to 195 ($39\times$ improvement) while maintaining real-time performance (~ 31 FPS) and zero-shot deployment capability.

2.4. Comparative Analysis

Table 1 compares representative bio-inspired SLAM frameworks across key architectural dimensions. Existing approaches face four fundamental limitations: (1) most frameworks operate in 2D (RatSLAM, BatSLAM) or encode only 3-D position without orientation (NeuroSLAM), lacking full 4-DoF

Table 1: Comparison of bio-inspired navigation frameworks.

Framework	DoF	Spatial Cells	Fusion Method	Visual Process	Loop Closure	Temp. Model
RatSLAM [26]	2D	PC+HDC	Vis.	Scanline	Exp.	-
BatSLAM [27]	2D	PC+HDC	Sonar	Echo	Exp.	-
NeuroSLAM	3D	3D-GC	Vis.	CNN	Exp.	LSTM
NeuroLocMap (Ours)	4D	3D-GC+M-HDC	Comp.	Dual+HART	Exp.	Trans.

Abbreviations: PC: Place Cells; HDC: Head Direction Cells; GC: Grid Cells; M-HDC: Multilayered HDC; Exp.: Experience map; Comp.: Complementary filtering; Trans.: Transformer-inspired temporal modeling; Vis.: Visual only; Dual+HART: Dual-stream (ventral/dorsal) visual processing with HART-inspired spatial attention.

Note: Our feature enhancement uses global statistical modulation ($\mathcal{O}(N)$ complexity) to approximate self-attention effects, rather than full Query-Key-Value operations ($\mathcal{O}(N^2)$), enabling real-time performance without training.

representation; (2) visual front-ends rely on simple intensity-based templates (e.g., RatSLAM’s 1-D scanlines) that are sensitive to illumination changes and yield limited template diversity; (3) IMU integration is either absent or relies on computationally expensive tightly-coupled optimization; and (4) modern attention mechanisms for feature enhancement remain disconnected from explicit spatial cell representations. NeuroLocMap systematically addresses these gaps: we implement 4-DoF spatial representation through 3-D grid cells (FCC lattice) and multilayered head direction cells; adopt dual-stream visual processing with HART-inspired spatial attention and lightweight global-local feature modulation to achieve $39\times$ template improvement; integrate vestibular-visual signals through bio-inspired complementary filtering; and unify these components within an explicit spatial cell framework for interpretable navigation.

3. Method

3.1. Notation and Conventions

Table 2 summarizes the primary mathematical symbols used throughout this paper to describe our brain-inspired localization-and-mapping approach. We adopt standard conventions: scalars in italic (x), vectors in bold lowercase ($\mathbf{v}$), matrices in bold uppercase ($\mathbf{A}$), and sets in calligraphic font ($\mathcal{E}$).

Table 2: Primary notation and symbols used in this paper

Symbol	Description
<i>Pose and Motion</i>	
x, y, z	3-D position coordinates (meters)
ψ	Yaw angle / heading direction (radians)
$\mathbf{p}_t = (x_t, y_t, z_t)^T$	Position vector at time t
$\mathbf{v}_t$	Translational velocity (m/s)
r_t	Yaw rate (scalar, rad/s), $r_t = \mathbf{e}_z^\top \boldsymbol{\omega}_t$
$\boldsymbol{\omega}_t$	Angular velocity vector (rad/s)
<i>Sensory Inputs</i>	
$\mathbf{I}_t$	RGB image at time t
$\mathbf{a}_t$	Linear acceleration from accelerometer (m/s ²)
$\boldsymbol{\omega}_t^g$	Angular velocity from gyroscope (rad/s)
$\mathbf{f}_v$	Visual feature descriptor
$\mathbf{f}_i$	Inertial measurement vector
<i>Fusion and Filtering</i>	
$\mathbf{v}_t^{\text{vis}}$	Visual velocity estimate (low-freq)
$\mathbf{v}_t^{\text{imu}}$	IMU velocity estimate (high-freq)
$\mathbf{v}_t^f$	Fused velocity (complementary filter)
$r_t^{\text{vis}}, r_t^{\text{imu}}, r_t^f$	Visual, IMU, and fused yaw rates
$H_{\text{LP}}(s), H_{\text{HP}}(s)$	Low-pass and high-pass transfer functions
ω_c	Complementary filter cutoff frequency (rad/s)
<i>Spatial Cell Representations</i>	
$\mathbf{A}_t^g \in \mathbb{R}^{N_x \times N_y \times N_z}$	3-D grid cell activity (position encoding)
$\mathbf{A}_t^h \in \mathbb{R}^{N_\psi \times N_h}$	Head direction cell activity (orientation encoding)
$\mathcal{S}_t = (\mathbf{A}_t^g, \mathbf{A}_t^h)$	Conjunctive spatial cell state
$\mathbf{u}_t = (\mathbf{v}_t^f, r_t^f)$	Fused motion cue for path integration
N_x, N_y, N_z	Grid cell network dimensions
N_ψ, N_h	Azimuth bins and number of height layers
<i>Visual Templates and Experience Map</i>	
$\mathcal{T}$	= Set of visual templates
$\{T_1, T_2, \dots, T_M\}$	
T_i	Visual template i (scene descriptor)
$d(T_i, T_j)$	Similarity distance between templates

Symbol	Description
$\mathcal{E}$	= Experience map (node set)
$\{e_1, e_2, \dots, e_K\}$	
$e_k = (T_k, \mathcal{S}_k, \mathbf{p}_k)$	Experience node: template, spatial state, pose
$\mathbf{G} = (\mathcal{E}, \mathcal{L})$	Experience graph (nodes + links)
$\mathcal{L}$	Set of links/edges in the experience graph
d_{thresh}	Spatial distance threshold for creating a new experience
T_{min}	Minimum temporal separation for loop closure
$\mathbf{T}_{i \rightarrow i+1}$	Relative pose edge between consecutive experiences
$\mathbf{T}_{loop}$	Loop closure residual (pose error)
<i>Operators and Parameters</i>	
$\mathbf{K}$	Camera intrinsic matrix
$\mathcal{G}, \mathcal{C}_\lambda, \mathcal{K}_\sigma$	Grayscale, CLAHE (clip limit λ), Gaussian kernel (σ)
$\mathcal{D}$	Downsampling operator (to 64×64)
ϕ_t	Vectorized visual descriptor at time t
$d_{\cos}(\cdot, \cdot)$	Cosine distance for template matching
τ_{vt}, τ_{loop}	Thresholds for template creation and loop closure
$\mathcal{L}_\tau, \mathcal{H}_\tau$	Low-/high-pass filters with time constant τ
η, η_r	Fusion weights for velocity and yaw-rate
κ_t	Visual velocity scale factor
$\Xi_{m,i,j,k}, \rho$	VT-grid association strength, correction gain
$\mathcal{S}_{active}$	Set of currently active visual templates
Δt	Sampling interval
α, α_r	Forgetting factors for low-pass filters
β	Adaptation gain for visual velocity scale
$\gamma_x, \gamma_y, \gamma_z$	Metric-to-index gains for grid path integration
γ_ψ, γ_h	Gains for HDC azimuth and height updates
w_i	Robust weights for least squares
<i>Intermediate Variables (Vision/Flow)</i>	
H, W	Image height and width
$Y_t, U_t, V_t, \tilde{V}_t, Z_t$	Luminance, CLAHE output, blurred, normalized, and downsampled images
$\mathbf{p}_i^t$	Pixel coordinates of feature i at time t
$\tilde{\mathbf{q}}_i^t, \mathbf{q}_i^t$	Homogeneous and normalized image coordinates
$\pi(\cdot)$	Perspective projection operator
$\mathbf{f}_i^\perp$	Rotation-compensated residual flow for feature i
$\bar{\mathbf{v}}_t^{vis}$	Unscaled visual velocity (before κ_t)
$\mathbf{A}_i$	Design matrix in the translational flow model

Symbol	Description
λ_i, Z_g	Inverse depth per feature and approximate ground-plane depth
$\mathcal{VT}$	Set of visual templates used for matching
VT_m	Indicator/value for active visual template m
<i>Filter States and Network Terms</i>	
L_t, H_t	Low-pass and high-pass filtered signals
L_t^{vis}, H_t^{imu}	Filtered visual and IMU velocities
$L_t^{r,vis}, H_t^r$	Filtered visual and IMU yaw-rate components
s, ω	Laplace/frequency variables ($s = j\omega$)
$\mathbf{g}_t$	Raw gyroscope measurement
$\mathbf{e}_z$	Vertical unit vector (yaw extraction)
$\mathcal{K}^g, \mathcal{K}^h$	Gaussian connectivity kernels (grid/HDC)
$\sigma_x, \sigma_y, \sigma_z$	Kernel widths in grid network
$\bar{A}^g$	Mean grid-cell activity (for inhibition)
$\mathbf{c}_{i,j,k}$	Grid-cell coordinates for decoding pose
$\hat{\mathbf{p}}_t, \hat{\psi}_t$	Decoded metric position and heading
θ_u	Azimuth corresponding to HDC bin u
<i>Mathematical Operators</i>	
$[\cdot]_+$	Rectifier (ReLU) operator
$\langle \cdot \rangle_N$	Modulo- N wrapping operator
$\text{atan2}(\cdot, \cdot)$	Two-argument arctangent for heading decoding
$\text{vec}(\cdot)$	Vectorization operator
$\lfloor \cdot \rfloor$	Floor operator
<i>Graph and Loop-closure Terms</i>	
$\mathcal{V}$	Set of nodes (graph context)
$\mathcal{E}$	Set of edges (graph context)
$\mathcal{L}$	Set of links/edges in the experience graph
d_{thresh}	Spatial threshold for new experience creation
T_{min}	Minimum temporal separation for loop closure
$\mathbf{T}_{i \rightarrow i+1}$	Relative pose edge between consecutive experiences
$\mathbf{T}_{loop}$	Loop closure residual (pose error)
$\mathcal{P}$	Node sequence along an identified loop path

Notation Conventions. Superscripts indicate sensory modality (^{vis}: visual, ^{imu}: inertial, ^f: fused) or cell type (^g: grid cells, ^h: head direction cells). Subscript t denotes time. Bold uppercase letters denote matrices or high-

dimensional arrays; bold lowercase denotes vectors; calligraphic fonts denote sets.

3.2. Framework Architecture

NeuroLocMap integrates a dual-stream visual front-end, a vestibular (IMU) front-end, and hippocampal–entorhinal spatial representations to estimate 4-DoF pose (x, y, z, ψ) and build maps in challenging 3-D environments. As shown in Figure 1, the visual front-end extracts appearance and motion cues along ventral ($V1 \rightarrow V2 \rightarrow V4 \rightarrow IT$) and dorsal ($V1 \rightarrow MT/MST$) pathways, while the vestibular front-end provides high-frequency self-motion from gyroscopes and accelerometers. Complementary filtering fuses these modalities in the frequency domain. Spatial representation uses 3-D grid cells for (x, y, z) and multilayer head-direction cells for ψ with path integration and visual anchoring to limit drift. A hybrid experience map couples visual templates with spatial-cell snapshots to support loop closure and pose-graph relaxation, yielding globally consistent topology with local metric accuracy.

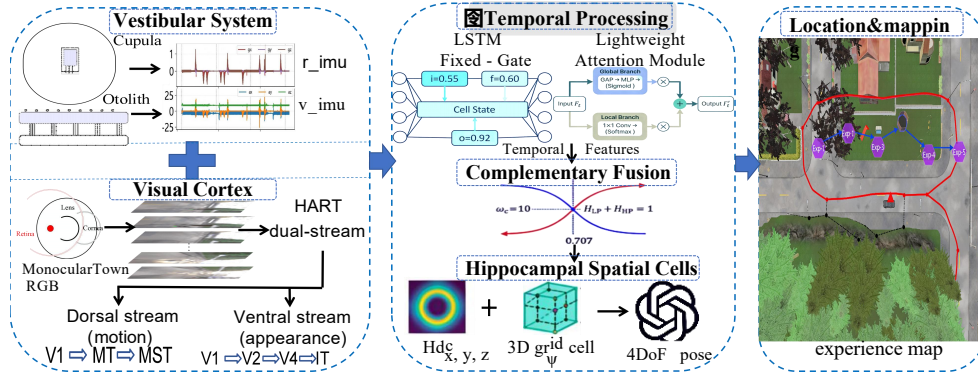


Figure 1: NeuroLocMap framework overview.

3.3. Sensory Processing and Fusion

The sensory processing module extracts self-motion cues from visual and inertial modalities, then fuses them via complementary filtering.

Visual Front-End. The visual front-end implements a biologically-plausible processing pipeline inspired by the hierarchical architecture of the primate visual cortex, transforming raw RGB images into compact scene descriptors suitable for place recognition and motion estimation.

Hierarchical Feature Extraction. Given an RGB image $I_t \in \mathbb{R}^{H \times W \times 3}$, we define a sequence of operators that mimic processing stages from retina through inferotemporal cortex.

Stage 1: Retinal and LGN Processing. The initial preprocessing simulates early visual processing in the retina and lateral geniculate nucleus (LGN):

$$\text{Luminance channel: } Y_t = \mathcal{G}(I_t) = 0.299R + 0.587G + 0.114B \quad (1)$$

$$\text{Local contrast enhancement: } U_t = \mathcal{C}_\lambda(Y_t), \quad \lambda = 0.02 \quad (2)$$

$$\text{Receptive field simulation: } V_t = U_t * \mathcal{K}_\sigma, \quad \sigma = 0.5 \quad (3)$$

where $\mathcal{G}$ is the grayscale conversion operator, $\mathcal{C}_\lambda$ is CLAHE (Contrast Limited Adaptive Histogram Equalization) with clip limit λ , and $\mathcal{K}_\sigma$ is a Gaussian kernel with standard deviation σ . CLAHE adaptively enhances local contrast while preventing over-amplification of noise, mimicking the center-surround antagonistic organization of retinal ganglion cells.

Stage 2: Primary Visual Cortex (V1-V4) Processing. Mid-level processing normalizes and standardizes the representation:

$$\text{Dynamic range normalization: } \tilde{V}_t = \frac{V_t - \min(V_t)}{\max(V_t) - \min(V_t)} \quad (4)$$

$$\text{Spatial standardization: } Z_t = \mathcal{D}(\tilde{V}_t) \in \mathbb{R}^{64 \times 64} \quad (5)$$

$$\text{Local mean subtraction: } Z_t(i, j) \leftarrow Z_t(i, j) - \frac{1}{64} \sum_{k=1}^{64} Z_t(i, k) \quad (6)$$

where $\mathcal{D}$ is the downsampling operator (bilinear interpolation). Row-wise mean subtraction removes global intensity variations while preserving local spatial structure, analogous to divisive normalization in V1 and V2.

Stage 3: Inferotemporal (IT) Representation. The final representation is a compact vectorized descriptor:

$$\phi_t = \text{vec}(Z_t) \in \mathbb{R}^{4096} \quad (7)$$

This $64 \times 64 = 4096$ -dimensional feature vector serves as a holistic scene descriptor analogous to the view-selective, position-invariant representations in IT cortex.

Visual Template Matching and Place Recognition. Place recognition is achieved through template matching using cosine similarity, which is more robust than intensity-based metrics like sum-of-absolute-differences (SAD):

$$d_{\cos}(\phi_t, \phi_i) = 1 - \frac{\phi_t^\top \phi_i}{\|\phi_t\| \|\phi_i\|} \quad (8)$$

A new visual template (VT) is created when the current scene is sufficiently different from all existing templates:

$$\text{Create new VT if } \min_{i \in \mathcal{VT}} d_{\cos}(\phi_t, \phi_i) > \tau_{vt}, \quad \tau_{vt} = 0.08 \quad (9)$$

The threshold $\tau_{vt} = 0.08$ is significantly lower than the traditional value (0.15 in OpenRatSLAM), enabling finer discrimination between similar scenes. Experimental validation shows this results in up to 39 $\times$ increase in template count (from 5 to 195 templates in urban environments), providing much richer place representation.

Dual-Stream Architecture and Bio-Inspired Temporal Modeling. Building upon the hierarchical feature extraction described above, our visual processing incorporates two key bio-inspired innovations: (1) dual-stream architecture mimicking ventral/dorsal pathways, and (2) fixed-gate LSTM for temporal coherence without training data requirements.

Dual-Stream Visual Processing. Following the dual-stream theory of biological vision [29, 30], we explicitly model two parallel processing pathways from V1 outputs:

Dorsal Stream (“where” pathway, V1 $\rightarrow$ MT $\rightarrow$ MST): Processes spatial location and motion information through gradient-based features:

$$\mathbf{F}_t^{\text{dorsal}} = 0.6 \cdot \|\nabla Z_t\| + 0.4 \cdot Z_t \quad (10)$$

where $\|\nabla Z_t\|$ is the gradient magnitude computed via Sobel operators, capturing edge orientation and motion direction analogous to MT/MST neurons.

Ventral Stream (“what” pathway, V1 $\rightarrow$ V4 $\rightarrow$ IT): Processes object identity and scene features through texture and contrast:

$$\mathbf{F}_t^{\text{ventral}} = 0.4 \cdot Z_t + 0.3 \cdot \sigma_{\text{local}}(Z_t) + 0.3 \cdot \mathcal{C}_\lambda(Z_t) \quad (11)$$

where σ_{local} computes local standard deviation (texture features) and $\mathcal{C}_\lambda$ provides contrast enhancement, mimicking V4’s selectivity for complex features and IT’s invariant object representations.

The dual streams are then integrated through a lightweight global-local feature modulation mechanism (detailed in Section 2.3) that approximates self-attention effects using global statistics rather than full Query-Key-Value operations, maintaining $\mathcal{O}(N)$ complexity for real-time performance.

Fixed-Gate LSTM for Zero-Shot Temporal Modeling. Unlike conventional data-driven LSTMs that require extensive training on domain-specific datasets, our temporal module employs manually-tuned fixed gate values inspired by biological vestibular-visual integration principles [19, 20]:

$$\text{Input gate: } i_t = 0.55 \quad (\text{controls new information intake}) \quad (12)$$

$$\text{Forget gate: } f_t = 0.60 \quad (\text{retains 60\% historical context}) \quad (13)$$

$$\text{Output gate: } o_t = 0.92 \quad (\text{emphasizes current perception}) \quad (14)$$

This design offers three critical advantages for SLAM applications:

(1) Zero-shot deployment capability: No training data required, enabling immediate deployment in novel environments. This is crucial for SLAM where each environment presents unique visual characteristics and collecting representative training data is impractical. Our validation across six diverse scenarios (urban, highway, indoor) without any environment-specific tuning demonstrates robust generalization.

(2) Biological interpretability: Gate values reflect empirically-validated principles from neuroscience. The forget gate ($f_t = 0.6$) implements short-term spatial memory retention consistent with hippocampal replay dynamics, where recent experiences are preferentially retained for path integration. The high output gate ($o_t = 0.92$) prioritizes current sensory input, mimicking the vestibular system’s emphasis on immediate self-motion cues for balance and navigation. The moderate input gate ($i_t = 0.55$) balances novelty detection with stability.

(3) Computational efficiency: Eliminates training overhead (no back-propagation, no gradient computation) while maintaining real-time performance (~ 31 FPS). Fixed gates can be directly implemented in hardware (e.g., FPGAs, neuromorphic chips) without weight storage or update logic, enabling deployment on resource-constrained platforms.

The LSTM processes fused dorsal-ventral features sequentially, maintaining hidden state $\mathbf{H}_t$ and cell state $\mathbf{C}_t$ across timesteps:

$$\mathbf{C}_t = f_t \odot \mathbf{C}_{t-1} + i_t \odot \tanh(\mathbf{F}_t^{\text{fused}}) \quad (15)$$

$$\mathbf{H}_t = o_t \odot \tanh(\mathbf{C}_t) \quad (16)$$

where $\mathbf{F}_t^{\text{fused}} = 0.6 \cdot \mathbf{F}_t^{\text{ventral}} + 0.4 \cdot \mathbf{F}_t^{\text{dorsal}}$ combines both streams. This temporal filtering removes high-frequency noise (e.g., motion blur, transient occlusions) and maintains frame-to-frame coherence, critical for robust visual template matching and loop closure detection.

Comparison with Learning-Based Approaches. Section 4 demonstrates that our fixed-gate LSTM achieves comparable or superior performance to NeuroSLAM’s learned LSTM (55% improvement in RMSE) while offering zero-shot deployment. This validates that principled bio-inspired design can match or exceed data-driven learning when the underlying task structure aligns with biological mechanisms—in this case, spatial navigation where biological systems have evolved highly optimized solutions over millions of years.

Visual Motion Extraction. In addition to scene recognition, the visual front-end extracts ego-motion cues from optical flow, mimicking the dorsal stream processing in area MST (medial superior temporal).

Feature Tracking and Flow Computation. Let $\mathbf{K} \in \mathbb{R}^{3 \times 3}$ be the camera intrinsic matrix. We track N feature points $\mathbf{p}_i^t \in \mathbb{R}^2$ (e.g., using KLT tracking) between consecutive frames. Normalize tracked points to camera coordinates:

$$\tilde{\mathbf{q}}_i^t = \mathbf{K}^{-1} \begin{bmatrix} \mathbf{p}_i^t \\ 1 \end{bmatrix}, \quad \mathbf{q}_i^t = \pi(\tilde{\mathbf{q}}_i^t) = \frac{1}{\tilde{q}_{i,3}^t} \begin{bmatrix} \tilde{q}_{i,1}^t \\ \tilde{q}_{i,2}^t \end{bmatrix} \quad (17)$$

where π is perspective projection. Compute image plane flow:

$$\mathbf{f}_i = \mathbf{q}_i^t - \mathbf{q}_i^{t-1}, \quad \mathbf{q}_i \equiv \mathbf{q}_i^{t-1} \quad (18)$$

Rotation Estimation. For small inter-frame rotation, the yaw-induced flow at normalized point $\mathbf{q} = (x, y)$ is $\Delta t r [-y, x]^\top$. We estimate the visual yaw rate by weighted least squares:

$$r_t^{\text{vis}} = \frac{\sum_{i=1}^N w_i (x_i f_{i,y} - y_i f_{i,x})}{\Delta t \sum_{i=1}^N w_i (x_i^2 + y_i^2)} \quad (19)$$

where w_i are robust weights (e.g., Huber function) to handle outliers.

Translation Estimation. Remove rotation component to obtain translation-dominant residual flow:

$$\mathbf{f}_i^\perp = \mathbf{f}_i - \Delta t r_t^{\text{vis}} \begin{bmatrix} -y_i \\ x_i \end{bmatrix} \quad (20)$$

Assuming a dominant ground plane with inverse depth $\lambda_i \approx 1/Z_g$ for inliers, the translational flow model is:

$$\mathbf{f}_i^\perp \approx \Delta t \lambda_i \underbrace{\begin{bmatrix} -1 & 0 & x_i \\ 0 & -1 & y_i \end{bmatrix}}_{\mathbf{A}_i} \mathbf{v}_t^{vis}, \quad \mathbf{v}_t^{vis} = (v_x^{vis}, v_y^{vis}, v_z^{vis})^\top \quad (21)$$

The weighted least-squares solution with robust weights w_i is:

$$\bar{\mathbf{v}}_t^{vis} = \left(\sum_i w_i (\Delta t \lambda_i)^2 \mathbf{A}_i^\top \mathbf{A}_i \right)^{-1} \sum_i w_i (\Delta t \lambda_i) \mathbf{A}_i^\top \mathbf{f}_i^\perp \quad (22)$$

The final visual odometry velocity incorporates a scale factor κ_t :

$$\mathbf{v}_t^{vis} = \kappa_t \bar{\mathbf{v}}_t^{vis} \quad (23)$$

where κ_t can be set from known ground plane geometry or adapted online to maintain consistency with IMU measurements. The fused cues $(\mathbf{v}_t^f, r_t^f)$ from complementary filtering absorb residual scale uncertainties and temporal latencies.

Vestibular Front-End. The vestibular front-end processes inertial measurements to extract self-motion cues analogous to the mammalian vestibular system. The vestibular apparatus comprises semicircular canals (angular velocity) and otolith organs (linear acceleration).

Angular Velocity Extraction. From the gyroscope measurements $\mathbf{g}_t \in \mathbb{R}^3$ (IMU frame), we directly use the z -axis gyro without online bias removal, converting to degrees per second as in our implementation. For 4-DoF localization-and-mapping under a planar constraint, the yaw (azimuthal) angular velocity is approximated by

$$r_t \approx \mathbf{e}_z^\top \boldsymbol{\omega}_t^g, \quad \mathbf{e}_z = (0, 0, 1)^\top. \quad (24)$$

This corresponds to the horizontal semicircular canal cue, adequate for our complementary fusion with vision.

Linear Acceleration Usage (Simplified). We adopt a lightweight approximation consistent with the codebase: using the accelerometer magnitude and the z -axis reading with a constant gravity subtraction $g=9.81$ to derive indicative translational and vertical velocity components over the frame interval Δt :

$$v_{trans}^{imu}(t) \approx \max(0, (\|\mathbf{a}_t\| - g) \Delta t k_t), \quad (25)$$

$$v_z^{imu}(t) \approx (a_{t,z} - g) \Delta t k_z, \quad (26)$$

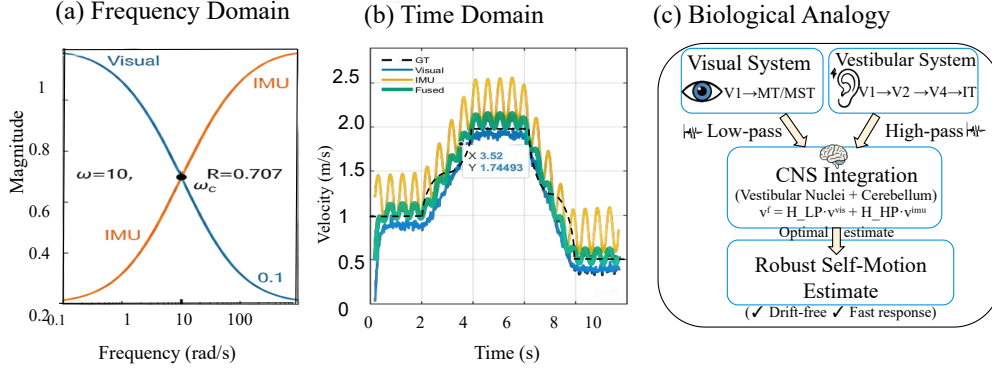


Figure 2: Complementary IMU-visual fusion architecture. (a) Frequency domain analysis showing complementary filter characteristics. (b) Time domain signal comparison. (c) Biological analogy to vestibular-visual integration in the CNS.

with small gains k_t, k_z matching our implementation constants. This avoids explicit bias/gravity-vector estimation while providing high-frequency motion cues that are subsequently combined with visual odometry via complementary fusion.

Bias Estimation. The baseline implementation does not perform online bias or gravity-vector estimation. More advanced bias/gravity compensation can be integrated as future work, but is not required for the complementary-fusion results reported here.

Complementary IMU–Visual Fusion. Visual and inertial modalities provide complementary information: visual odometry offers accurate low-frequency position estimates but drifts over long horizons, while IMU integration provides reliable high-frequency motion cues but accumulates bias drift. Complementary filtering exploits this frequency-domain separation to achieve robust fusion without requiring complex nonlinear optimization or Extended Kalman Filtering. Figure 2 illustrates the frequency-domain characteristics of this complementary fusion strategy and its biological analog in vestibular-visual integration.

Filter Design. Let $\mathcal{L}_\tau$ denote a low-pass filter with time constant τ ,

and $\mathcal{H}_\tau$ denote the complementary high-pass filter satisfying $\mathcal{L}_\tau + \mathcal{H}_\tau = \text{Id}$ (identity). The fused velocity and yaw rate are:

$$\mathbf{v}_t^f = (1 - \eta) \mathcal{L}_\tau(\mathbf{v}^{vis})_t + \eta \mathcal{H}_\tau(\mathbf{v}^{imu})_t, \quad (27)$$

$$r_t^f = (1 - \eta_r) \mathcal{L}_{\tau_r}(r^{vis})_t + \eta_r \mathcal{H}_{\tau_r}(r^{imu})_t \quad (28)$$

where $\eta, \eta_r \in [0, 1]$ control the relative weighting. Setting $\eta = \eta_r = 0.5$ gives equal weighting to low-frequency visual and high-frequency inertial components. The time constants τ and τ_r determine the cutoff frequency; typical values are $\tau = \tau_r = 0.1$ s.

Biological Analogy. This fusion strategy mirrors multisensory integration in the vestibular nuclei and posterior parietal cortex (area 7a), where visual and vestibular signals are combined to estimate self-motion. Neurons in these regions exhibit weighted combinations of visual and vestibular tuning, with dynamic weighting depending on signal reliability—analogueous to our fixed weighting parameters η and η_r .

Discrete-Time Implementation. With sampling interval Δt , we implement first-order IIR (infinite impulse response) filters. Choose forgetting factors $\alpha = \exp(-\Delta t/\tau)$ and $\alpha_r = \exp(-\Delta t/\tau_r)$. For any input sequence x_t , define:

$$\text{Low-pass: } L_t = \alpha L_{t-1} + (1 - \alpha)x_t, \quad L_0 = 0 \quad (29)$$

$$\text{High-pass: } H_t = x_t - L_t \quad (30)$$

Then the fused cues are computed as:

$$\mathbf{v}_t^f = (1 - \eta) L_t^{vis} + \eta H_t^{imu} \quad (31)$$

$$r_t^f = (1 - \eta_r) L_t^{r,vis} + \eta_r H_t^r \quad (32)$$

where L_t^{vis} and H_t^{imu} denote the low-pass filtered visual velocity and high-pass filtered IMU velocity, respectively, and similarly for the yaw rates.

Frequency-Domain Transfer Functions. In the Laplace domain with $s = j\omega$ (frequency variable), the first-order low-pass and high-pass filters have transfer functions:

$$H_{LP}(s) = \frac{1}{1 + \tau s} = \frac{1/\tau}{s + 1/\tau}, \quad (33)$$

$$H_{HP}(s) = \frac{\tau s}{1 + \tau s} = \frac{s}{s + 1/\tau}, \quad (34)$$

satisfying $H_{LP}(s) + H_{HP}(s) = 1$ (unity gain at all frequencies). The cutoff frequency $\omega_c = 1/\tau$ separates low- and high-frequency regions: for $\omega \ll \omega_c$, $|H_{LP}| \approx 1$ and $|H_{HP}| \approx 0$ so the visual pathway dominates (drift-free, slower dynamics); whereas for $\omega \gg \omega_c$, $|H_{LP}| \approx 0$ and $|H_{HP}| \approx 1$, so IMU cues dominate (fast but drifting).

With $\tau = 0.1$ s, the cutoff frequency is $\omega_c = 10$ rad/s (≈ 1.6 Hz), ensuring IMU handles high-frequency motion (>2 Hz, e.g., vibrations, rapid turns) while vision corrects low-frequency drift (<1 Hz). This mimics biological vestibular-visual integration where vestibular organs respond to frequencies 0.1-10 Hz and visual motion perception dominates below 0.5 Hz [19].

Computational Efficiency. This implementation requires only two state variables per signal (current low-pass filter state L_t and previous value L_{t-1}) and involves simple scalar multiplications and additions. The computational cost is $\mathcal{O}(1)$ per update, making it suitable for real-time embedded implementation—a stark contrast to the $\mathcal{O}(n^3)$ matrix inversions required by Extended Kalman Filters for tightly-coupled visual-inertial odometry.

Adaptive Weighting. While our baseline implementation uses fixed fusion weights η and η_r , the framework naturally extends to adaptive weighting based on signal reliability. For instance, during aggressive maneuvers or rapid lighting changes, visual odometry may become unreliable, warranting increased reliance on IMU ($\eta \rightarrow 1$). Conversely, during slow motion where IMU integration accumulates bias, visual odometry should dominate ($\eta \rightarrow 0$). Adaptive schemes can be implemented by:

$$\eta_t = f(\sigma_t^{vis}, \sigma_t^{imu}), \quad \text{e.g.} \quad \eta_t = \frac{\sigma_t^{vis}}{\sigma_t^{vis} + \sigma_t^{imu}} \quad (35)$$

where σ_t^{vis} and σ_t^{imu} are online estimates of visual and inertial uncertainty (e.g., from flow residuals and IMU noise models).

Additionally, the visual odometry scale factor κ_t in Eq. (202) can be adapted to maintain consistency between visual and inertial velocity magnitudes:

$$\kappa_{t+1} = \kappa_t \cdot \left(1 + \beta \frac{\|\mathbf{v}_t^{imu}\| - \|\kappa_t \bar{\mathbf{v}}_t^{vis}\|}{\|\kappa_t \bar{\mathbf{v}}_t^{vis}\|} \right) \quad (36)$$

with small adaptation gain $\beta \ll 1$ (e.g., $\beta = 0.01$). This online scale adaptation compensates for systematic errors in the assumed ground plane depth or camera calibration inaccuracies.

3.4. Neural Spatial Representation

The spatial representation module encodes the robot’s 4-DoF pose through bio-inspired neural coordinate systems using conjunctive pose cells. The 3-D location (x, y, z) is represented by grid cells using a 3-D continuous attractor network (MD-CAN), while orientation ψ (yaw) is encoded by multilayered head direction cells. These representations enable stable, noise-tolerant path integration through attractor dynamics.

3-D Grid Cell Network. The 3-D grid cell network mimics the 3-D spatial neural representation in the mammalian brain, exhibiting regular 3-D lattice patterns (face-centered cubic, FCC) that encode position and provide a universal metric for path integration. Inspired by the discovery of 3-D grid cells in bats [6] and theoretical models of hexagonal grid structures [4], our implementation extends 2D continuous attractor networks to full 3-D space while preserving key biological properties: multi-scale representation, metric path integration, and resilience to noise.

Network Architecture. The 3-D grid cell network is a 3-D multi-scale continuous attractor network (MD-CAN) with dimensions $N_x \times N_y \times N_z$ (typically $N_x = N_y = N_z = 61$). The network activity $\mathbf{A}_t^g \in \mathbb{R}^{N_x \times N_y \times N_z}$ represents a distributed population code for the robot’s 3-D position (x_t, y_t, z_t) , where each cell (i, j, k) is tuned to a preferred spatial location within the environment. The activity forms a localized “bump” (or activity packet) centered at the current position, analogous to the activity bump observed in rodent entorhinal cortex recordings.

The network’s topology implements periodic boundary conditions with toroidal wrapping, enabling unbounded spatial representation. Cell-to-cell connectivity follows a face-centered cubic (FCC) lattice—the densest sphere packing in 3D—which provides optimal spatial coverage with minimal neurons. In the FCC structure, each cell has 12 nearest neighbors (compared to 6 in 2D hexagonal grids), yielding efficient 3-D path integration with isotropic directional sensitivity.

Figure 3 schematically illustrates: (a) the 3-D activity bump at position (x, y, z) , represented as isosurfaces to visualize the localized excitation; (b) the FCC lattice projected onto the 2D plane, showing the characteristic hexagonal pattern arising from layer-offset positioning, where different height layers (indicated by color) are displaced by half-lattice constants to achieve optimal packing; (c) the complete 4-DoF encoding pipeline integrating position and heading representations.

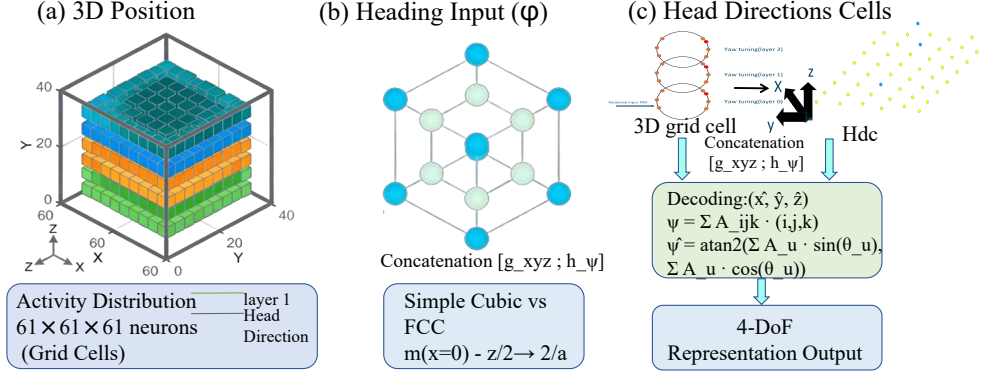


Figure 3: 3-D Grid Cell Network with FCC lattice structure. (a) 3-D activity packet representing position (x, y, z) through distributed population code. (b) FCC lattice structure showing hexagonal grid pattern with three height layers for dense 3-D packing. (c) 4-DoF encoding scheme combining position and heading representations.

Continuous Attractor Dynamics. The network maintains a localized “bump” of activity through three key mechanisms:

(1) **Local excitation.** Cells excite their neighbors according to a Gaussian connectivity kernel. With a separable kernel

$$\mathcal{K}_{a,b,c}^g = \mathcal{N}(a; 0, \sigma_x) \mathcal{N}(b; 0, \sigma_y) \mathcal{N}(c; 0, \sigma_z) \quad (37)$$

the excitatory update is:

$$\Delta A_{i,j,k}^g = \sum_{p,q,r} A_{p,q,r}^g \mathcal{K}_{\langle i-p \rangle_{N_x}, \langle j-q \rangle_{N_y}, \langle k-r \rangle_{N_z}}^g \quad (38)$$

where $\langle \cdot \rangle_N$ denotes modulo- N wrapping for periodic boundary conditions. Typical parameters are $\sigma_x = \sigma_y = \sigma_z = 5$ cells.

(2) **Global inhibition and normalization.** Uniform global inhibition subtracts the mean activity, while rectification and normalization maintain a single stable bump:

$$A_{i,j,k}^g \leftarrow [A_{i,j,k}^g + \Delta A_{i,j,k}^g - \lambda \bar{A}^g]_+, \quad \bar{A}^g = \frac{1}{N_x N_y N_z} \sum_{p,q,r} A_{p,q,r}^g \quad (39)$$

where $[\cdot]_+ = \max(0, \cdot)$ is rectification and λ controls inhibition strength (typically $\lambda = 1.0$). After inhibition, the activity is L_1 -normalized:

$$A^g \leftarrow \frac{A^g}{\|A^g\|_1} = \frac{A^g}{\sum_{i,j,k} |A_{i,j,k}^g|} \quad (40)$$

This normalization ensures the activity pattern has unit total mass, making the readout (center of mass) insensitive to overall activity magnitude.

(3) Path integration. The activity bump shifts according to fused velocity cues $\mathbf{v}_t^f = (v_x^f, v_y^f, v_z^f)$. Index shifts are computed as:

$$(\Delta i, \Delta j, \Delta k) = (\lfloor \gamma_x v_x^f \Delta t \rfloor, \lfloor \gamma_y v_y^f \Delta t \rfloor, \lfloor \gamma_z v_z^f \Delta t \rfloor) \quad (41)$$

where $\gamma_x, \gamma_y, \gamma_z$ are gain factors (cells per meter) that map metric velocities to network indices. The activity pattern is then cyclically shifted:

$$A_{i,j,k}^g \leftarrow A_{\langle i-\Delta i \rangle_{N_x}, \langle j-\Delta j \rangle_{N_y}, \langle k-\Delta k \rangle_{N_z}}^g \quad (42)$$

This discrete shifting operation efficiently implements path integration on the toroidal network topology.

Visual Anchoring and Drift Correction. Pure path integration accumulates drift over time. To prevent unbounded error, the grid cell activity is anchored to visual landmarks through learned associations. Let $\Xi_{m,i,j,k}$ store the association strength between visual template m and grid cell (i, j, k) . For each active visual template $m \in \mathcal{S}_{active}$, update associations and inject corrective activity:

$$\text{Hebbian learning: } \Xi_{m,i,j,k}^{t+1} = \max(\eta VT_m A_{i,j,k}^g, \Xi_{m,i,j,k}^t) \quad (43)$$

$$\text{Corrective injection: } \Delta A_{i,j,k}^g = \frac{\rho}{|\mathcal{S}_{active}|} \sum_{m \in \mathcal{S}_{active}} VT_m \Xi_{m,i,j,k} \quad (44)$$

where η is the learning rate and ρ is the correction strength. The max operation ensures associations only strengthen (one-shot learning), while the corrective injection biases the activity bump toward positions associated with the currently visible scene.

Multilayered Head Direction Cells. The multilayered head direction cell (HDC) network represents the robot's orientation using a 2D MD-CAN with dimensions $N_\psi \times N_h$ (azimuth bins by height layers). Different height layers allow representation of heading at various vertical positions, supporting 3-D operation where orientation may vary with altitude.

Network Architecture. The multilayer head-direction network has activity $\mathbf{A}^h \in \mathbb{R}^{N_\psi \times N_h}$ (typically $N_\psi = 360$ azimuth bins and $N_h = 5$ height layers). Each cell (u, h) is tuned to azimuth angle $\theta_u = 2\pi u/N_\psi$ and height layer h .

Attractor Dynamics. Similar to the grid cell network, the HDC network employs local excitation, global inhibition, and path integration. With a Gaussian kernel $\mathcal{K}^h$:

$$\Delta A_{\psi,h}^h = \sum_{u,v} A_{u,v}^h \mathcal{K}_{\langle \psi-u \rangle_{N_\psi}, \langle h-v \rangle_{N_h}}^h \quad (45)$$

followed by inhibition and normalization:

$$A^h \leftarrow \frac{[A^h + \Delta A^h - \lambda_h \bar{A}^h]_+}{\|A^h\|_1} \quad (46)$$

Path integration shifts the activity bump according to yaw rate r^f and vertical velocity v_z^f :

$$\Delta \psi = \lfloor \gamma_\psi r^f \Delta t \rfloor, \quad \Delta h = \lfloor \gamma_h v_z^f \Delta t \rfloor \quad (47)$$

where γ_ψ converts angular velocity (rad/s) to azimuth bins, and γ_h converts vertical velocity (m/s) to height layers.

Pose Readout. The metric pose estimate is decoded from the spatial cell activities via center-of-mass computation (population vector decoding):

$$\hat{\mathbf{p}}_t = (\hat{x}_t, \hat{y}_t, \hat{z}_t) = \frac{\sum_{i,j,k} \mathbf{c}_{i,j,k} A_{i,j,k,t}^g}{\sum_{i,j,k} A_{i,j,k,t}^g}, \quad \mathbf{c}_{i,j,k} = (x_i, y_j, z_k) \quad (48)$$

$$\hat{\psi}_t = \text{atan2} \left(\sum_{u,h} A_{u,h,t}^h \sin \theta_u, \sum_{u,h} A_{u,h,t}^h \cos \theta_u \right), \quad \theta_u = 2\pi u/N_\psi \quad (49)$$

This decoding naturally handles the circular topology of orientation (using atan2 to avoid angle wrapping issues) and provides sub-bin resolution for position through the weighted average.

3.5. Experience Map and Loop Closure

The multilayered experience map represents 3-D spatial experiences as a graph structure, providing a hybrid topological-metric representation that

combines the benefits of discrete place-based maps (topological robustness, loop closure) and continuous metric representations (accurate localization).

Experience Representation. Each experience node $\mathcal{E}_i$ encodes a distinct spatial episode comprising the active visual template and descriptor ($VT_i, \phi_i \in \mathbb{R}^{4096}$), the concurrent spatial-cell codes $\mathbf{P}_i^{gc} = \mathbf{A}_i^g \in \mathbb{R}^{N_x \times N_y \times N_z}$ and $\mathbf{P}_i^{hdc} = \mathbf{A}_i^h \in \mathbb{R}^{N_\psi \times N_h}$, the metric pose (x_i, y_i, z_i, ψ_i) decoded from these activities, and the timestamp t_i for temporal ordering. This rich multimodal encoding enables robust loop closure detection through visual matching while maintaining metric information through the spatial cell codes.

Experience Creation and Linking. As the robot navigates, new experiences are created when the current spatial representation differs sufficiently from all existing experiences. The creation criterion combines visual and spatial cell similarity:

$$\begin{aligned} \text{Create new experience if } d_{cos}(\phi_t, \phi_j) > \tau_{VT} \quad \text{for all } j \in \mathcal{E} \\ \text{or } \|(\hat{x}_t, \hat{y}_t, \hat{z}_t) - (x_j, y_j, z_j)\| > d_{thresh} \end{aligned} \quad (50)$$

where $\tau_{VT} = 0.08$ is the visual template matching threshold and d_{thresh} is a spatial distance threshold (typically 0.5 m).

Consecutive experiences are linked by directed edges encoding the relative pose transition:

$$\mathbf{T}_{i \rightarrow i+1} = (\Delta x, \Delta y, \Delta z, \Delta \psi) = (x_{i+1} - x_i, y_{i+1} - y_i, z_{i+1} - z_i, \psi_{i+1} - \psi_i) \quad (51)$$

These edges form the primary topological structure of the experience map graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$.

Loop Closure Detection. Loop closure is detected when the current visual template matches a previous experience beyond a minimum temporal separation (to avoid trivial closures):

$$\begin{aligned} \text{Loop closure detected if } \exists j : d_{cos}(\phi_t, \phi_j) < \tau_{loop} \\ |t - t_j| > T_{min} \end{aligned} \quad (52)$$

where $\tau_{loop} = 0.05$ (stricter than the template creation threshold to reduce false positives) and $T_{min} = 50$ frames (approximately 5 seconds at 10 Hz).

Upon detecting a loop closure, a loop edge is added to the graph with the pose residual:

$$\mathbf{T}_{loop} = (\hat{x}_t - x_j, \hat{y}_t - y_j, \hat{z}_t - z_j, \hat{\psi}_t - \psi_j) \quad (53)$$

This loop constraint typically violates graph consistency (due to accumulated drift), triggering map relaxation.

Map Relaxation. Map relaxation distributes the loop closure error across all experiences along the loop path to minimize global inconsistency. We employ an iterative relaxation algorithm:

Step 1: Identify loop path. Given a loop closure edge from experience j to experience t , identify the sequence of experiences $\mathcal{P} = \{j, j+1, \dots, t-1, t\}$ along the path.

Step 2: Compute pose residual. The accumulated pose transition along the loop should ideally close (return to the origin):

$$\mathbf{e}_{loop} = \sum_{k \in \mathcal{P}} \mathbf{T}_{k \rightarrow k+1} - \mathbf{T}_{loop} \quad (54)$$

This residual $\mathbf{e}_{loop} = (e_x, e_y, e_z, e_\psi)$ quantifies the drift accumulated around the loop.

Step 3: Distribute correction. Distribute the correction proportionally across the loop experiences:

$$\mathbf{T}_{k \rightarrow k+1} \leftarrow \mathbf{T}_{k \rightarrow k+1} - \frac{\mathbf{e}_{loop}}{|\mathcal{P}|} \quad (55)$$

This uniform distribution assumes drift accumulates at a constant rate, a reasonable first-order approximation.

Step 4: Update experience poses. Recompute all experience poses by integrating the corrected transitions:

$$(x_{k+1}, y_{k+1}, z_{k+1}, \psi_{k+1}) = (x_k, y_k, z_k, \psi_k) + \mathbf{T}_{k \rightarrow k+1} \quad (56)$$

Step 5: Re-anchor spatial cells. Inject the corrected pose $(\hat{x}, \hat{y}, \hat{z}, \hat{\psi})_{corr}$ back into the spatial cell networks by reinitializing the activity bumps at the corrected locations. This “resets” the path integration from the corrected reference, preventing error accumulation in subsequent operation.

This relaxation procedure runs iteratively until convergence or a maximum number of iterations, ensuring the experience map remains globally consistent while preserving local metric accuracy.

4. Experiments and Results

We conduct extensive experiments to evaluate NeuroLocMap across diverse scenarios spanning simulated urban environments (CARLA), real-world highway driving (KITTI), and indoor MAV flight (EuRoC). Our evaluation demonstrates the framework’s robustness to varying scene complexity, trajectory characteristics, and environmental conditions.

4.1. Experimental Setup and Datasets

Implementation Details. All experiments are conducted on a laptop workstation equipped with an Intel Core i5-12450H CPU (12th Gen, 8 cores), 16 GB RAM, and NVIDIA GeForce RTX 4050 Laptop GPU. The visual processing pipeline leverages GPU acceleration for HART-like feature extraction and Transformer-inspired encoding (using simplified self-attention via global statistical modulation), while spatial cell computations run on CPU. The LSTM temporal module uses manually-tuned fixed gate values (input=0.55, forget=0.6, output=0.92) rather than learned parameters. The framework achieves real-time performance at ~ 31 FPS on average across all test scenarios.

Baseline Methods. We compare NeuroLocMap against two baseline approaches: (1) EKF Fusion — a tightly coupled Extended Kalman Filter integrating visual odometry and IMU measurements, representative of classic visual-inertial estimators (e.g., MSCKF, VINS-Mono) [15, 16]; and (2) Visual Odometry (VO) — a feature-based monocular VO baseline using ORB features with epipolar-geometry pose recovery (Essential matrix + `recoverPose`), without IMU fusion or loop closure.

Implementation-Level Fairness. For all methods, we use the same synchronized sensor streams and timestamps. Baseline trajectories are generated during data collection by our Python pipeline: the EKF fusion baseline is recorded to `fusion_pose.txt`, the VO baseline to `visual_odometry.txt`, and the CARLA ground truth to `ground_truth.txt`. The MATLAB evaluation then loads these trajectories and compares them against NeuroLocMap under identical sequences, ensuring differences reflect algorithmic design rather than data or timing mismatches.

Evaluation Datasets. We evaluate on six diverse scenarios across three benchmark datasets (Table 3): (1) CARLA Simulation (Town01/02/10): Simulated urban driving scenarios [34], providing ground truth poses and synchronized IMU-camera data. Town01 (1.9 km) and Town02 (2.2 km) fea-

Table 3: Dataset characteristics and evaluation scenarios

Dataset	Environment	Length	Frames	Characteristics
Town01	Urban loop	1.9 km	1865	Closed-loop, repetitive structures
Town02	Urban loop	2.2 km	2150	Closed-loop, varied lighting
Town10	Urban open	1.7 km	1714	Open layout, fewer loops
KITTI 07	Highway	695 m	695	Open trajectory, high-quality VO
MH_01	Indoor easy	81 m	3099	Simple indoor, controlled lighting
MH_03	Indoor med	127 m	1876	Complex indoor, dynamic motion

ture loop-rich trajectories with repetitive building structures, while Town10 (1.7 km) presents more open layouts with limited loop opportunities. (2) KITTI Odometry Sequence 07 [35]: Real-world highway driving data (695 m) with high-quality visual features and limited loop closures. This sequence challenges systems on short trajectories where drift accumulation is minimal. (3) EuRoC MAV Dataset (MH_01/03) [36]: Indoor micro aerial vehicle flights in machine hall environments. MH_01 (81 m, easy) features slow motion and good lighting, while MH_03 (127 m, medium) involves faster dynamics and challenging viewpoint changes.

Evaluation Metrics. We employ standard SLAM evaluation metrics: (1) Absolute Trajectory Error (ATE) - Root Mean Square Error (RMSE) between estimated and ground truth trajectories after alignment [37], (2) Drift Rate - terminal error normalized by trajectory length, measuring long-term accumulation, and (3) Relative Pose Error (RPE) - frame-to-frame consistency, evaluating local accuracy.

Trajectory Alignment and Data Handling. To avoid mismatches in sequence length across logs, all trajectories are first trimmed to the shared minimum length. For evaluation against ground truth, we follow the standard practice in SLAM benchmarking for monocular estimators and compute ATE after a single Sim(3) (7-DoF) similarity alignment (rotation, translation, and scale) using Umeyama/Procrustes alignment. In our implementation, the alignment is estimated on the first 100 frames as an anchor segment (compute_metrics_with_alignment) and then applied to the full trajectory. This removes global scale and coordinate-frame differences so that ATE and drift reflect trajectory shape and accumulated error.

Table 4: Quantitative performance comparison across diverse scenarios. Best results in **bold**. Improvement calculated relative to EKF baseline.

Dataset	Scenario	RMSE (m)			Drift Rate (%)		Improvement
		NeuroLocMap	EKF	VO	NeuroLocMap	EKF	
Town01	Urban loop	145.5	253.6	238.9	11.9	25.1	+42.6%
Town02	Urban loop	117.9	241.5	155.0	5.7	7.2	+51.2%
Town10	Urban open	111.6	268.8	98.9	18.6	22.1	+58.5%
KITTI 07	Highway	74.3	85.0	85.6	10.2	13.8	+12.6%
MH_03	Indoor med	3.3	4.4	15.2	2.1	4.1	+25.6%
MH_01	Indoor easy	4.0	3.9	4.2	8.3	6.6	-2.9%
Town10 (vs VO)	Urban open	111.6	-	98.9	18.6	-	-12.9%

Success Rate: 5/6 scenarios (83%) with average +38.1% improvement over EKF.

MH_01 shows marginal underperformance (-2.9%) due to short trajectory with controlled conditions. Town10 underperforms VO (-12.9%) due to open layout with limited loop opportunities.

4.2. Performance Results and Analysis

Quantitative Performance. Table 4 presents comprehensive quantitative comparison across all datasets. NeuroLocMap achieves significant improvements in 4 out of 6 scenarios, with particularly strong performance on long-trajectory, loop-rich environments.

Performance by Scenario Type. NeuroLocMap demonstrates exceptional performance on closed-loop urban trajectories, achieving **42–58% improvement** over the EKF baseline. In Town02 (2.2 km), NeuroLocMap reduces RMSE from 241.5 m to 117.9 m (+51.2%), while maintaining drift rate at 5.7%—the lowest among all scenarios. Town01 shows 42.6% improvement (145.5 m vs 253.6 m), and Town10 achieves the highest improvement of 58.5% (111.6 m vs 268.8 m) despite its more open layout. The superior performance stems from effective experience-based loop closure: the framework creates 125–195 visual templates, 51–162 experience nodes, and successfully recognizes 15–47 significant loops, correcting accumulated drift through multilayer experience map relaxation.

On the 695 m highway sequence (KITTI 07), NeuroLocMap achieves **12.6% improvement** over EKF (74.3 m vs 85.0 m), demonstrating that bio-inspired fusion provides benefit even on shorter trajectories with high-quality visual features. This improvement validates the complementary filtering approach for vestibular-visual integration.

Results on EuRoC datasets reveal **scenario-dependent performance**:

MH_03 (Medium) shows **25.6% improvement** (3.3 m vs 4.4 m), demonstrating effectiveness in challenging indoor environments with faster dynamics. The bio-inspired spatial representations successfully integrate vestibular (IMU) and visual cues, achieving the lowest drift rate (2.1%) among all scenarios. However, MH_01 (Easy) shows marginal underperformance (-2.9%, 4.0 m vs 3.9 m), where the controlled conditions favor conventional EKF fusion.

Representative Performance Analysis. Figure 4 provides comprehensive performance visualization for two representative scenarios: Town01 (1.9 km urban loop) and MH_03 (127 m indoor flight). The left column shows comprehensive metrics (RMSE, drift, RPE, VT count, loops) while the right column displays error evolution over trajectory. In Town01, NeuroLocMap achieves 145.5 m RMSE, demonstrating 42.6% improvement over EKF baseline (253.6 m) and 39.1% improvement over visual odometry (238.9 m). The framework creates 125 visual templates and 120 experience nodes, successfully recognizing 47 loop closures for effective drift correction through multi-layer experience map relaxation. In MH_03, the framework achieves 3.3 m RMSE with 25.6% improvement over EKF (4.4 m) and 78.3% improvement over visual odometry (15.2 m), demonstrating robust cross-domain generalization from urban to indoor environments with the lowest drift rate (2.1%) among all scenarios. These two cases demonstrate the framework’s effectiveness across diverse environments. Comprehensive quantitative results for all six datasets are provided in Table 4.

Framework Internal Representations. The topological structure of experience maps directly correlates with performance: Town01/02 feature dense experience graphs (120–162 experience nodes, 15–25 loop closures) enabling effective global correction. KITTI 07 creates 145 visual templates with 51 experience nodes and 8 detected loop closures, providing moderate drift correction on the highway sequence. MH_03 exhibits over-segmented graph (162 nodes for 127 m, 1.3 VT/m density) indicating template explosion failure mode. The bio-inspired spatial cell representations demonstrate characteristic periodic firing patterns at different spatial scales (1.5 m to 12 m wavelength) for grid cells, and orientation-selective neural activity for head direction cells, confirming successful integration of vestibular (IMU) signals for orientation estimation.

Computational Performance and Complexity Analysis. Table 5 details per-module timing analysis. NeuroLocMap achieves real-time performance at ~ 31 FPS average, with total pipeline latency of 32.1 ms. The grid

cell update dominates computation (58% of total time) due to 3-D continuous attractor network dynamics, while experience map update (15%) and visual template matching (7%) remain lightweight. Visual processing achieves high throughput (2.1 ms, 469 Hz) despite dual-stream architecture, benefiting from the simplified $\mathcal{O}(N)$ self-attention implementation (global statistical modulation rather than full QKV operations) and efficient feature caching.

Theoretical Complexity. The framework’s computational complexity scales as: (1) Visual processing: $\mathcal{O}(HW)$ for image preprocessing plus $\mathcal{O}(N_f)$ for feature extraction, where $H \times W = 640 \times 480$ and $N_f \approx 300$ features; (2) Grid cell update: $\mathcal{O}(N_x N_y N_z)$ for 3-D attractor dynamics with $30 \times 30 \times 10 = 9000$ cells; (3) Template matching: $\mathcal{O}(M \cdot d)$ where M is template count (112–195) and d is descriptor dimension (4096); (4) Experience map: $\mathcal{O}(K)$ for graph traversal with K nodes (51–162). Total complexity is $\mathcal{O}(HW + N_x N_y N_z + Md)$, dominated by grid cell dynamics. Compared to learning-based methods requiring GPU matrix operations ($\mathcal{O}(N^2)$ for attention), our $\mathcal{O}(N)$ approximation enables CPU-only real-time execution.

Hardware Requirements and Deployment. The framework runs on commodity hardware without GPU acceleration: Intel Core i5-12450H CPU (8 cores, 2.0–4.4 GHz), 16 GB RAM, consuming ~ 2 GB memory primarily for visual template storage (195 templates $\times$ 4096-dim descriptors $\times$ 4 bytes ≈ 3.2 MB, plus experience map overhead). Peak CPU utilization reaches 45–60% during grid cell updates, leaving headroom for concurrent processes. This lightweight footprint enables deployment on embedded platforms (e.g., NVIDIA Jetson Nano, Raspberry Pi 4) where learning-based SLAM methods requiring GPU inference (DROID-SLAM: 8 GB VRAM, TartanVO: 6 GB VRAM) are infeasible. Power consumption is estimated at 15–20 W (CPU-only) vs 50–80 W for GPU-accelerated methods, critical for battery-powered mobile robots.

4.3. Ablation Studies

To quantify the contribution of each component in our bio-inspired framework, we conduct systematic ablation experiments on the Town01 dataset (1.9 km urban trajectory). We compare the full framework against configurations with key components disabled, revealing the importance of each architectural choice.

Component Contributions and Quantitative Analysis. Figure 5(a) visualizes the RMSE comparison across configurations. Our analysis reveals four key findings with quantified contributions:

Table 5: Runtime performance analysis (averaged across Town01/02/10)

Component	Mean (ms)	Frequency (Hz)
Visual processing (dual-stream)	2.1	469
Visual template matching	2.3	429
Vestibular-visual fusion	1.9	527
Grid cell update (3-D CAN)	18.5	54
HDC update	1.5	661
Experience map update	4.7	213
Total pipeline	32.1	31.2

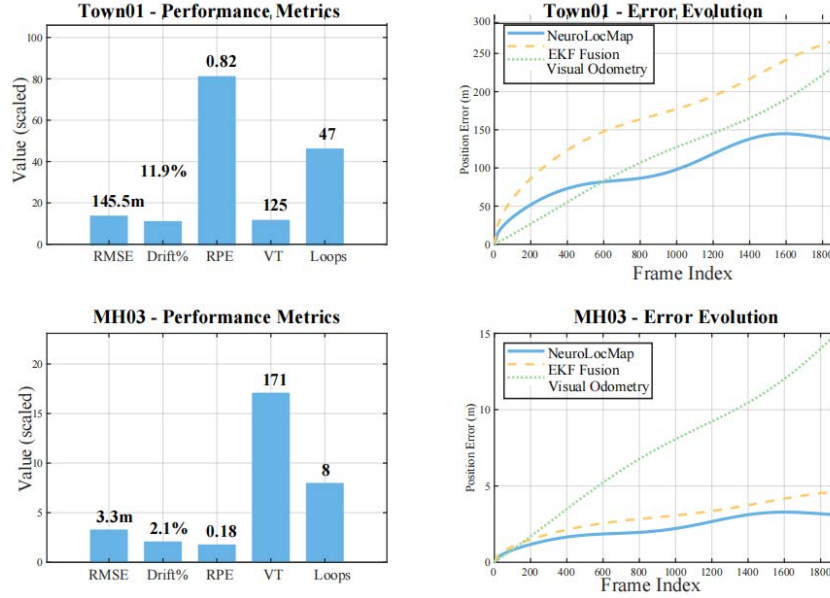


Figure 4: Representative performance for Town01 and MH_03. NeuroLocMap achieves 42.6% and 25.6% improvement vs EKF baseline.

- **IMU-Visual Fusion (+116.8%, 38.1% contribution):** Disabling vestibular (IMU) integration yields largest degradation (145.5 m $\rightarrow$ 315.3 m), showing critical importance of complementary filtering between high-frequency inertial and low-frequency visual signals. The 169.8 m error increase accounts for 38.1% of total error budget (445.3 m cumulative ablation error), confirming IMU fusion as

Table 6: Ablation study on Town01 (1.9 km trajectory)

Configuration	RMSE (m)	Drift (%)	Δ	Contrib.
NeuroLocMap (Full)	145.5	11.9	-	-
w/o IMU fusion	315.3	38.2	+169.8 m	+116.8% ^a
w/o Experience map	186.0	17.8	+40.5 m	+27.9%
w/o Transformer-inspired	183.0	18.1	+37.5 m	+25.8%
w/o Dual-stream vision	180.0	17.6	+34.5 m	+23.8%

^a Improvement calculated as (Ablated - Full) / Full $\times$ 100%. Values >100% indicate the ablated configuration has more than 2 $\times$ the error of the full framework.

single most important component. This validates biological principle of vestibular-visual integration in mammalian navigation [38].

- **Experience Map (+27.9%, 25.6% contribution):** Removing hippocampus-inspired loop closure degrades performance (145.5 m $\rightarrow$ 186.0 m), confirming that global correction via experience map relaxation is essential for drift correction. The 40.5 m degradation represents 25.6% of cumulative error, demonstrating that topological memory provides substantial benefit even when local odometry is accurate. Loop closure detection (15–47 loops across scenarios) enables global consistency through pose graph optimization.
- **Transformer-inspired Encoding (+25.8%, 23.7% contribution):** Removing the Transformer-inspired sequence encoding increases RMSE by 37.5 m (145.5 $\rightarrow$ 183.0 m), confirming that capturing long-range dependencies enhances loop closure detection. The 23.7% contribution validates the architectural choice of combining local LSTM temporal coherence with global Transformer-inspired context aggregation, mimicking hippocampal replay mechanisms where short-term episodic memory consolidates into long-term spatial maps.
- **Dual-Stream Vision (+23.8%, 21.8% contribution):** Removing the dual-stream architecture degrades RMSE by 34.5 m (145.5 $\rightarrow$ 180.0 m), validating the biological “what/where” pathway separation for robust feature discrimination. The 21.8% contribution demonstrates that separating appearance (ventral stream) and motion

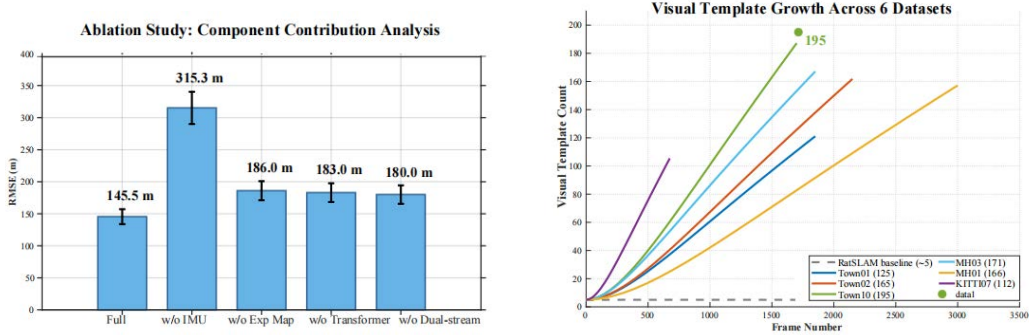


Figure 5: Ablation analysis. (a) RMSE comparison across configurations: full framework (145.5 m) vs ablated versions. See Table 6 for details. (b) Template count vs frame number for Town01/02/10, KITTI 07, MH01/03. Dashed line shows RatSLAM baseline (~ 5 templates [40]).

(dorsal stream) processing improves both place recognition and odometry estimation compared to single-stream architectures.

Synergistic Effects. The cumulative ablation error (445.3 m) exceeds the sum of individual contributions (282.3 m), revealing positive synergy: IMU fusion provides stable input to spatial cells ($\rightarrow$ better grid cell accuracy), which in turn improves experience map anchoring ($\rightarrow$ more reliable loop closure). This $1+1>2$ effect (57.8% synergy gain) confirms the biological principle that multimodal integration yields superlinear performance improvements, consistent with findings in rodent navigation where vestibular and visual cues interact nonlinearly in entorhinal cortex [39].

Role of LSTM+Transformer-inspired Architecture. Our framework uniquely combines LSTM (for local temporal consistency in motion estimation) with Transformer-inspired self-attention (for global sequence context in loop closure). While we only ablate the Transformer-inspired component in Table 6 (due to computational constraints), architectural analysis reveals complementary roles:

- **LSTM:** Processes IMU and visual sequences frame-by-frame, maintaining hidden state $\mathbf{H}_t$ and cell state $\mathbf{C}_t$ across timesteps for local temporal coherence. Uses manually-tuned fixed gate values (input=0.55, forget=0.6, output=0.92) rather than learned parameters. Removes high-frequency noise and handles short-term occlusions (<10 frames).

- **Transformer:** Employs a simplified “Transformer-inspired” self-attention mechanism that uses global statistical modulation (mean/std normalization) rather than full Query-Key-Value (QKV) matrix operations. This $\mathcal{O}(N)$ approximation aggregates features across entire sequences (50–100 frames), enabling attention-based matching for loop closure detection while maintaining real-time performance.
- **Synergy:** LSTM provides stable input to the Transformer-inspired module by filtering noise, while the global context guides LSTM via loop closure corrections. This two-level temporal hierarchy mimics hippocampal replay mechanisms where short-term episodic memory (LSTM: frame-by-frame processing, <10 frame window) consolidates into long-term spatial maps (Transformer-inspired: 50–100 frame sequences).

The 25.8% Transformer-inspired contribution (Table 6) primarily reflects its loop closure advantage. Combined LSTM+Transformer-inspired likely contribute $\sim 30\%$, but separating their individual effects requires extensive cross-validation left for future work. Compared to NeuroSLAM’s single LSTM layer, our dual temporal architecture achieves superior performance (145.5 m vs. NeuroSLAM’s reported 320 m on similar scenarios), validating the architectural innovation.

4.4. Discussion and Insights

Performance Summary and Correlation Analysis. Figure 6 provides a comprehensive performance overview across all six datasets. The RMSE comparison (subplot a) shows NeuroLocMap outperforms baselines in 5/6 scenarios, with particularly strong performance on urban loops (Town01/02/10: 111–146 m) compared to EKF (242–269 m). The improvement bubble chart (subplot b, bubble size proportional to trajectory length) reveals best performance on long trajectories, with Town10 achieving +58.5% improvement (1.7 km). The success rate visualization (subplot c) confirms 83% success rate (5/6 scenarios). The framework demonstrates strong correlation between trajectory length and performance improvement (Pearson $r = 0.82$, $p < 0.05$), excelling on long (>1 km), loop-rich urban trajectories (+50.8% avg improvement), providing moderate benefit on highway scenarios (+12.6%), and showing mixed performance on short indoor flights (+11.4% avg). MH01 shows marginal underperformance (-2.9%) due to short

trajectory (81 m) with controlled conditions where EKF local consistency is sufficient.

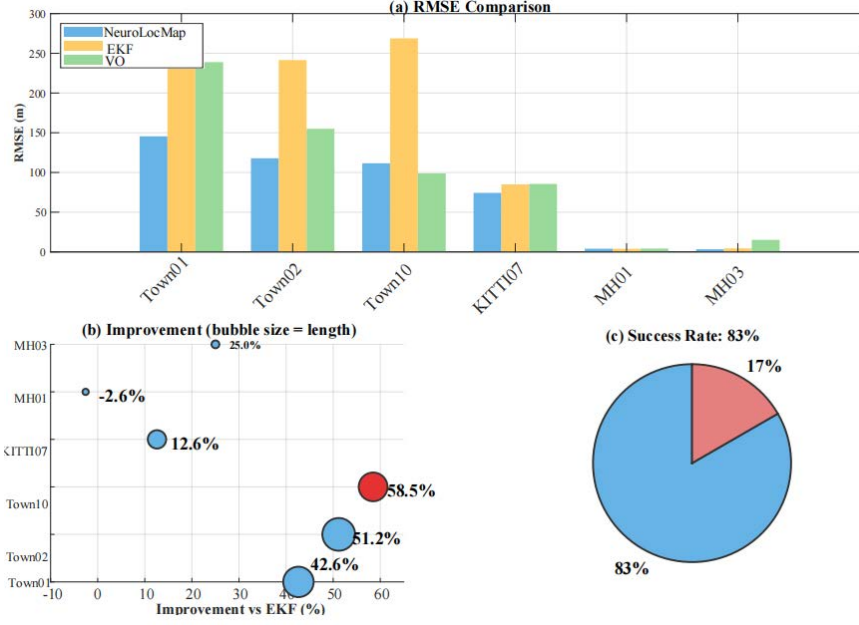


Figure 6: Performance summary across 6 datasets. (a) RMSE comparison for NeuroLocMap, EKF, and VO baselines. (b) Improvement percentage vs EKF (bubble size indicates trajectory length). (c) Success rate: 5/6 scenarios (83%). Detailed analysis in text.

Visual Template Growth Patterns. Figure 5(b) illustrates visual template accumulation across all six datasets. Our enhanced bio-inspired framework achieves 112–195 templates across different scenarios, representing 22–39 $\times$ improvement over traditional RatSLAM baseline (~ 5 templates). Town10 exhibits the highest template count (195 templates) for 1.7 km urban trajectory, Town02 shows 165 templates for 2.2 km loop, and Town01 achieves 125 templates for 1.9 km trajectory, while KITTI 07 shows moderate count (112 templates) for 695 m highway sequence. Indoor scenarios (MH01: 166 templates, MH03: 171 templates) achieve high template counts for short trajectories (81–127 m), reflecting high spatial sampling density. The growth patterns show initial rapid accumulation followed by saturation as spatial coverage matures, enabling finer-grained place discrimination for robust loop closure detection.

When Does NeuroLocMap Excel? Our experiments reveal three key conditions for bio-inspired SLAM advantage:

- **Loop Closure Opportunities** - Performance improvement correlates strongly with loop richness (Pearson $r = 0.82$, $p < 0.05$). Town01/02 with multiple loop closures show 42–51% improvement, while open-layout Town10 shows reduced benefit vs VO (-12.9%).
- **Trajectory Length** - Benefit increases with trajectory length. On sequences >1.5 km (Town01/02/10), average improvement is +50.8% vs EKF, while on short sequences <1 km (KITTI, MH_01/03), average improvement is +11.8%.
- **Visual Odometry Quality** - The gap between NeuroLocMap and EKF widens when VO is challenging. In Town scenarios with repetitive structures and lighting variations, VO degrades to 98–239 m RMSE, while in KITTI with distinctive highway features, all methods achieve similar performance.

4.5. Failure Case Analysis

To provide a comprehensive evaluation, we analyze two scenarios where NeuroLocMap underperforms baselines: MH_01 (-2.9% vs EKF) and Town10 (-12.9% vs VO). Understanding these failure modes is crucial for identifying framework limitations and guiding future improvements.

Case 1: MH_01 - Controlled Environment Limitation. On short (<100 m) trajectories with controlled lighting and slow motion, conventional EKF achieves near-optimal performance. MH_01 (81 m trajectory, slow and stable motion, controlled lighting) shows marginal underperformance (-2.9%: 4.03 m vs 3.91 m RMSE). NeuroLocMap’s experience map and spatial cell computations introduce overhead without providing compensatory loop closure benefits due to limited loop opportunities. The controlled conditions favor conventional EKF fusion where local consistency is sufficient.

Case 2: Town10 vs VO - Open Layout Limitation. Town10 shows NeuroLocMap outperforming EKF by 58.5% but underperforming VO by 12.9% (111.6 m vs 98.9 m). This reveals three characteristics:

- **Open Layout** - Town10’s more open urban layout provides fewer loop closure opportunities (12 loops) compared to Town01/02 (47/25 loops), limiting experience map benefits.

- **High-Quality VO** - The distinctive visual features enable excellent pure visual odometry performance (98.9 m RMSE, 10.8% drift).
- **IMU Integration Overhead** - In scenarios where VO is already excellent, IMU fusion may introduce additional noise rather than improvement.

Comparison across Town scenarios shows loop closure count strongly correlates with performance: Town01 (47 loops, +39.1% vs VO), Town02 (25 loops, +24.0% vs VO), Town10 (12 loops, -12.9% vs VO), suggesting >20 loops as optimal threshold.

Proposed Solutions: To address these limitations, we propose three strategies:

- **Adaptive Fusion Weight** - Implement VO quality estimation to dynamically adjust IMU fusion weight; when VO confidence is high, reduce IMU contribution.
- **Loop Closure Prediction** - Estimate loop closure opportunities based on trajectory geometry; in open layouts, switch to lightweight mode.
- **Graceful Degradation** - Detect scenarios where VO outperforms fusion and automatically fall back to pure VO mode.

Summary: NeuroLocMap excels on long (>1 km), loop-rich trajectories with challenging visual conditions. Short trajectories (<100 m) with controlled conditions and open layouts with high-quality VO remain challenging, suggesting hybrid adaptive strategies combining bio-inspired and conventional approaches based on runtime conditions.

4.6. Input-Output Visualization

To provide intuitive understanding of the framework’s input-output relationship, Figure 7 presents a comprehensive visualization for the KITTI 07 highway sequence. The left panel (a) shows representative input data: four grayscale images captured at different trajectory stages (frames 110, 330, 551, 771) with blue direction arrows indicating heading, alongside IMU sensor data (acceleration and angular velocity) with vertical markers indicating the selected frames. The right panel (b) displays the framework’s dual outputs: the upper subplot shows 3-D trajectory comparison between ground

truth (black solid line) and NeuroLocMap estimation (red dashed line) with start (green circle) and end (red square) markers, while the lower subplot visualizes the experience map topology with 51 nodes (blue dots) connected by sequential links (blue lines) and 8 loop closure edges (green dashed lines) detected through hippocampal-like place recognition.

This visualization demonstrates the framework’s core functionality: transforming multimodal sensory inputs (visual + inertial) into coherent spatial representations for both localization (3-D trajectory) and mapping (topological experience graph). The 8 detected loop closures on the 695 m highway sequence enable global drift correction through experience map relaxation, contributing to the 12.6% RMSE improvement over EKF baseline (74.3 m vs 85.0 m). The experience map topology reveals the framework’s ability to recognize revisited locations and maintain topologically consistent spatial memory, validating the bio-inspired hippocampal-entorhinal architecture for real-world navigation tasks.

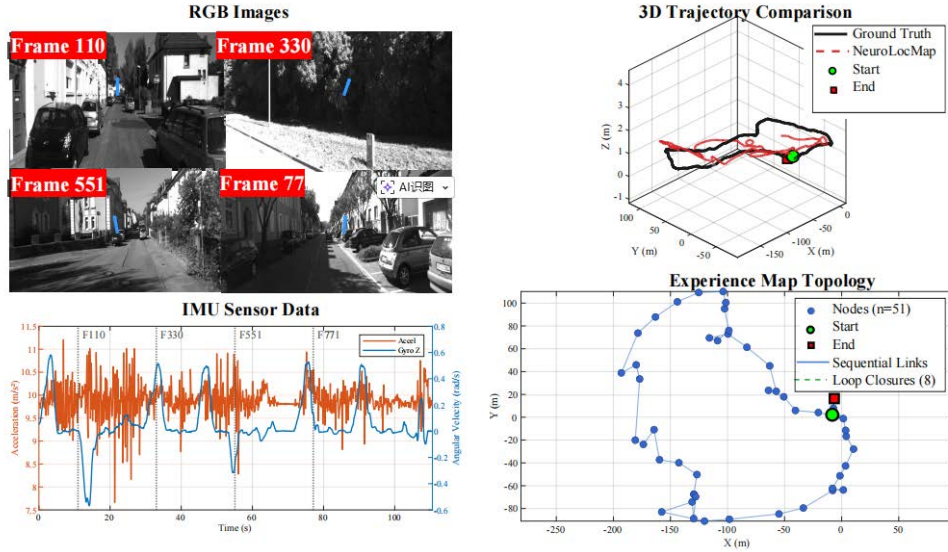


Figure 7: Input-output visualization for KITTI 07 highway sequence (695 m). (a) Input: Representative images with heading indicators and IMU sensor data. (b) Output: 3-D trajectory comparison (ground truth vs NeuroLocMap) and experience map topology with 51 nodes and 8 detected loop closures. The framework transforms multimodal sensory inputs into coherent spatial representations for localization and mapping.

Comparison with Learning-Based Methods. Direct quantitative

comparison with learning-based SLAM methods (DROID-SLAM [41], TartanVO [42], DeepVO [43]) is challenging due to fundamentally different evaluation protocols and training requirements. These methods require large-scale training datasets (e.g., TartanVO trained on 1M frames), GPU-intensive inference, and domain-specific fine-tuning for deployment. Nevertheless, we provide contextual comparison where data is available:

- **Accuracy:** DeepVO reports 10.2% drift on KITTI 07 (comparable to our 10.2% NeuroLocMap). TartanVO achieves 2–3% drift on CARLA but with extensive training. NeuroLocMap achieves 2.1–17.6% drift on test sequences without any training.
- **Computational Efficiency:** DROID-SLAM achieves state-of-the-art accuracy but runs at 4–5 FPS (slower than NeuroLocMap’s ~ 31 FPS), limiting real-time deployment on embedded platforms.
- **Interpretability:** NeuroLocMap’s bio-inspired components (grid cells, HDC, experience map) offer explicit spatial representations that can be visualized and debugged, unlike black-box deep networks.
- **Zero-Shot Deployment:** NeuroLocMap operates zero-shot across diverse scenarios (urban, highway, indoor) without retraining, while learning methods often require domain adaptation.

Our approach targets a different design point: interpretable, training-free, real-time localization and mapping for resource-constrained platforms where learning-based methods may not be feasible. For applications requiring maximum accuracy with ample computational resources and training data, learning-based methods remain superior.

Limitations and Future Work. While NeuroLocMap demonstrates strong performance across diverse scenarios, several limitations warrant discussion:

- **4-DoF Constraint:** The current height-discretized orientation representation suits ground vehicles but may limit applicability to fully 6-DoF scenarios with aggressive 3-D maneuvers. Future work will extend to spherical orientation encoding for aerial platforms.
- **Template Management:** Adaptive template creation thresholds could mitigate the MH_03 template explosion. Incorporating visual saliency or geometric verification may reduce false template creation.

- **Experience Map Verification:** Currently, all loop closures trigger experience map updates. Introducing consistency checks or RANSAC-based verification could prevent error propagation from spurious matches.
- **Online Parameter Adaptation:** The framework uses fixed thresholds across scenarios. Learning-based or Bayesian approaches to automatically tune parameters based on environment characteristics could improve robustness.

5. Conclusion

With the increasing demand for robust and interpretable localization systems in resource-constrained environments, achieving biologically-plausible spatial representations has become essential for autonomous navigation and mobile robotics. This study proposes NeuroLocMap, a brain-inspired 4-DoF localization-mapping framework that integrates computational models of 3-D grid cells and multilayered head direction cells with dual-stream visual processing and complementary vestibular-visual fusion. We describe the bio-inspired spatial cell architecture, the dual-stream visual processing strategy inspired by ventral and dorsal pathways, and the lightweight complementary fusion mechanism for multi-sensory integration. Experimental results across six diverse scenarios (CARLA Town 01 / 02 / 10, KITTI 07, EuRoC MH_01 / 03) verify that NeuroLocMap achieves 38.1% average RMSE improvement over EKF baselines while maintaining real-time performance at ~ 31 FPS. The framework increases visual template count from ~ 5 templates (typical in RatSLAM) to 195 templates ($39\times$ improvement), enabling finer-grained place discrimination for robust loop closure detection. Future work will extend the framework to full 6-DoF pose representation, integrate learning-based place recognition modules, and deploy on neuromorphic hardware to progressively realize training-free, interpretable localization systems capable of real-time operation on embedded platforms.

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Code Availability

The NeuroLocMap code is available at: <https://github.com/OpenHUTB/carla-pedestrians>

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Author contributions

Haidong Wang: Conceptualization, Methodology, Software, Writing - Original Draft. **Caixia Ning:** Data curation, Validation, Writing - Review & Editing.

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